

Entangling two atoms with one global laser pulse

Challenge 01 — solved, cloud-validated, run on the real quantum computer

Fidelity 0.9926 / 0.7500 (baseline) → 1.000000 / 1.000000 (ours)

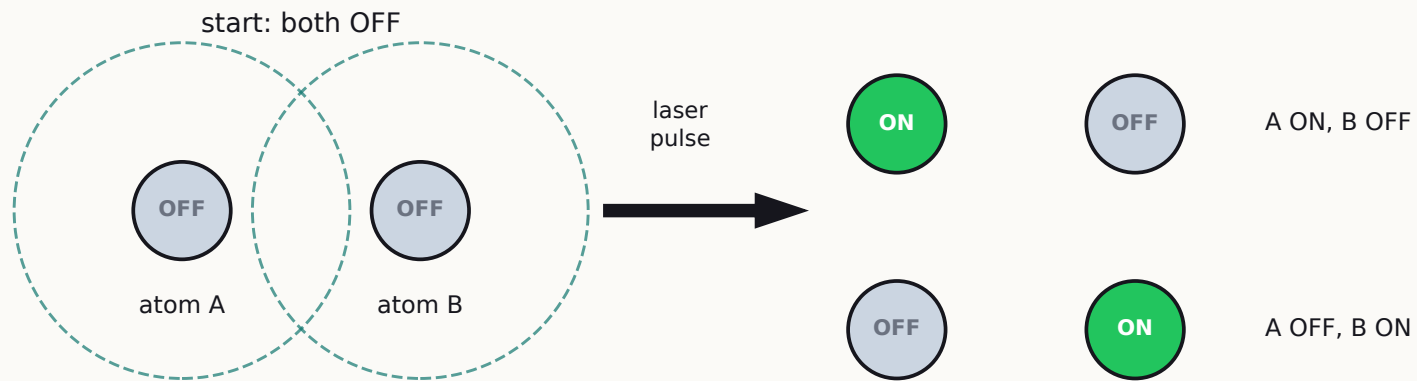
Real atoms: 89.4% of 500 shots entangled

The problem, in one slide

- Two atoms, 5.0 or 6.5 micrometers apart. One laser shines on both — we cannot address them individually.
- Goal state: $|\Psi^+\rangle = (|gr\rangle + |rg\rangle)/\sqrt{2}$ — exactly one atom excited, shared between both. That is entanglement.
- Our only controls: laser power $\Omega(t)$ and laser frequency offset $\delta(t)$, over at most 6000 ns.
- Score: $F = |\langle\Psi^+|\psi(T)\rangle|^2$ — how close the final state is to the goal (1.0 = perfect).
- Success = beat the starter pulse at BOTH spacings, inside the machine's published limits.

Physics we exploit: two nearby excited atoms repel ($V = C_6/r^6$). Inside the blockade radius $R_b \approx 7.2 \mu\text{m}$, double excitation is forbidden — that forbidden-ness is what creates entanglement.

The idea in one picture



dashed circles: the blockade —
circles overlap, so BOTH ON
is physically forbidden

BOTH at the same time — that is entanglement

(like flipping two coins and getting HT + TH together)

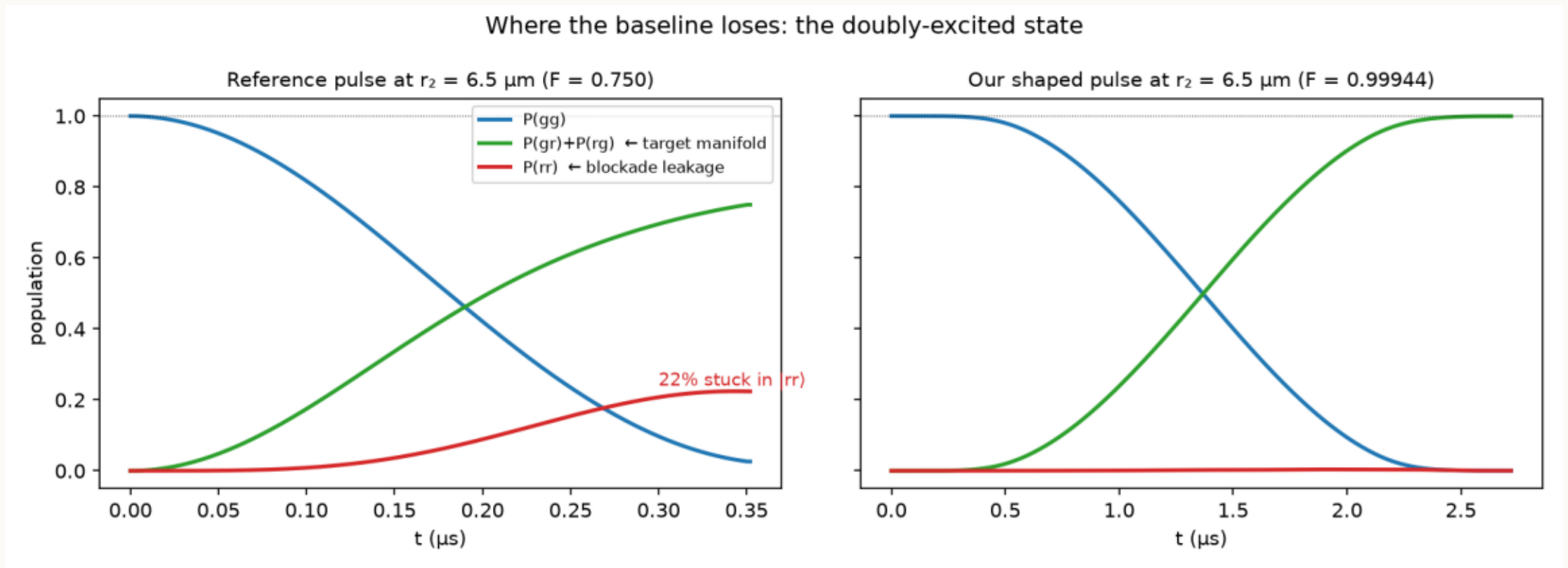
- An atom is a light switch:
 $|g\rangle = \text{OFF}$, $|r\rangle = \text{ON}$.
- Close atoms cannot both be ON
(the blockade). So the pulse can
only put ONE excitation in —
shared between the two atoms.
- That shared, both-options-at-once
state IS the Bell state we're scored on.

The starter pulse works at one spacing, fails at the other

- Baseline: constant power $\Omega = 2\pi \times 1$ MHz for 352 ns, no frequency shaping.
- At $r_1 = 5.0$ μm the blockade is strong ($V/\Omega = 8.8$): $F = 0.9926$. Fine.
- At $r_2 = 6.5$ μm the blockade is weak ($V/\Omega = 1.83$): $F = 0.7500$. Broken.
- Where does 25% go? Into the forbidden $|rr\rangle$ state — 22% leaks straight through the weak blockade.

**One number — V/Ω , interaction over drive — explains the whole gap.
 V is fixed by the atom spacing. But Ω is OURS to choose.**

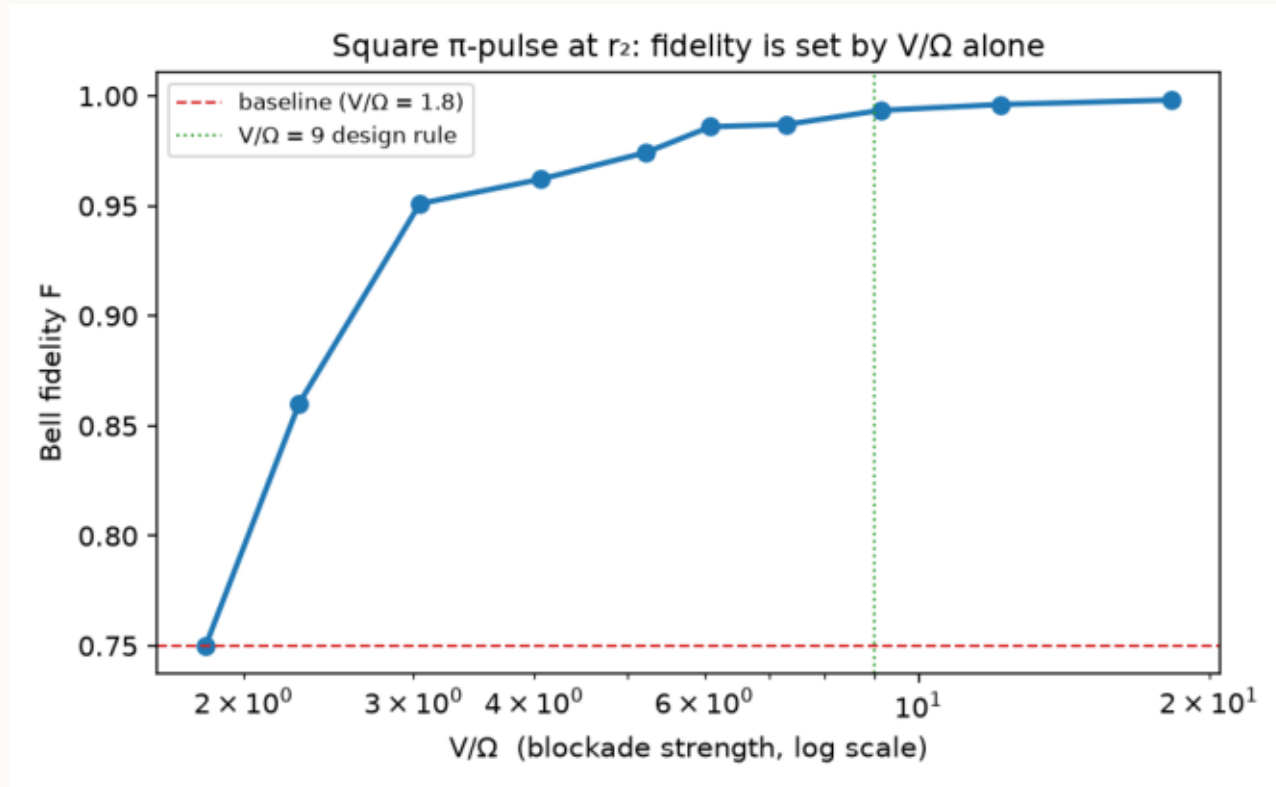
Watch the baseline fail — then watch ours not



x-axis: time during the pulse (μs) · y-axis: probability of each atomic configuration

- Left (baseline): the red curve is the forbidden $|rr\rangle$ state filling up to 22% — the green target stalls at 0.75.
- Right (ours): red stays at zero for the whole pulse; the green target reaches 1.0.

The fix is one knob: slow down to restore the blockade



- Same square pulse, only Ω lowered (pulse lengthened to match).

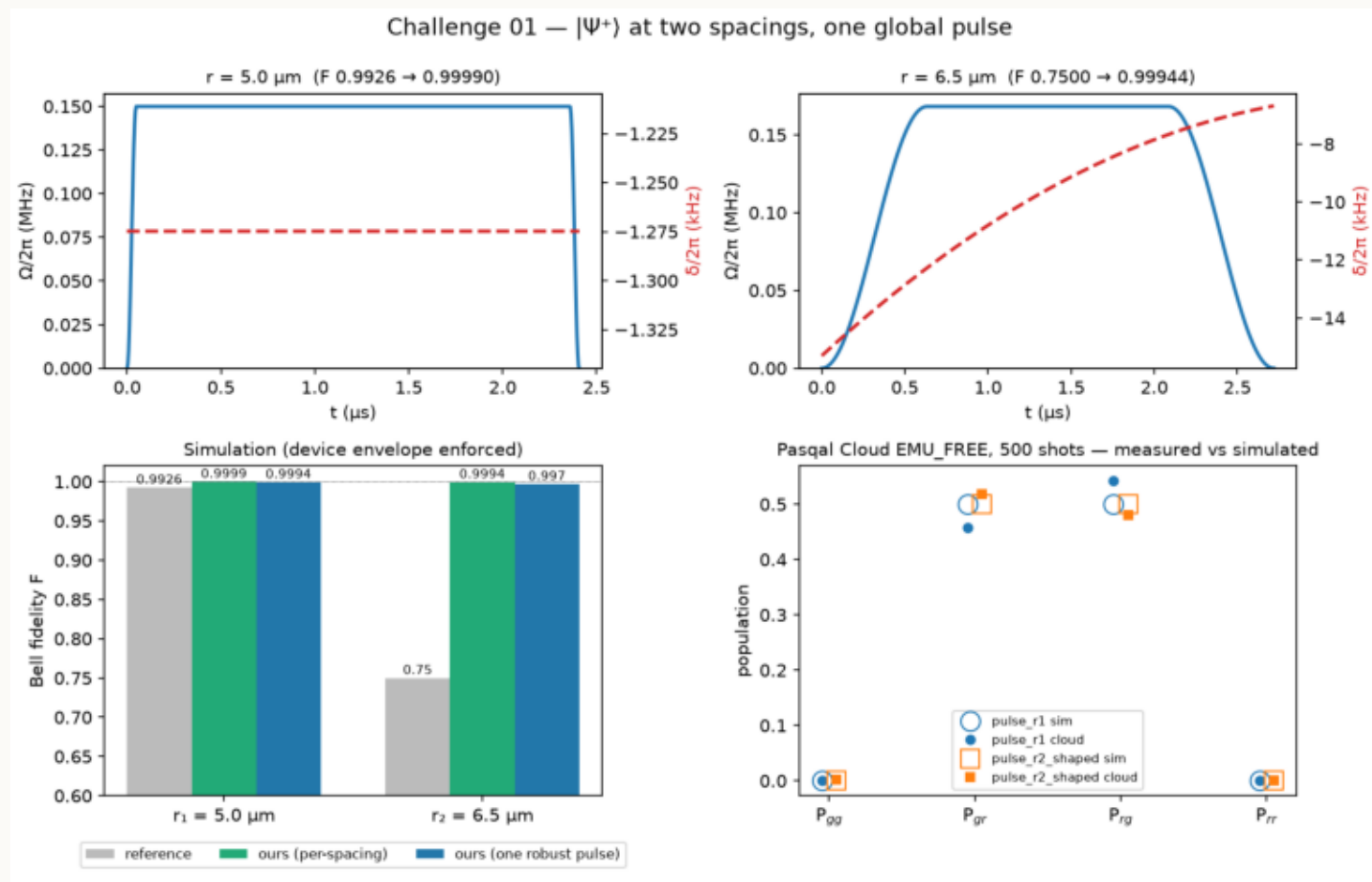
- F climbs monotonically with V/Ω and crosses 0.99 at $V/\Omega \approx 9$.

- That became our design rule — and the 6000 ns budget makes it free: the baseline used only 352 ns.

- On top: smooth ramps + a tiny δ that cancels the energy shift $\Omega^2/2V$.

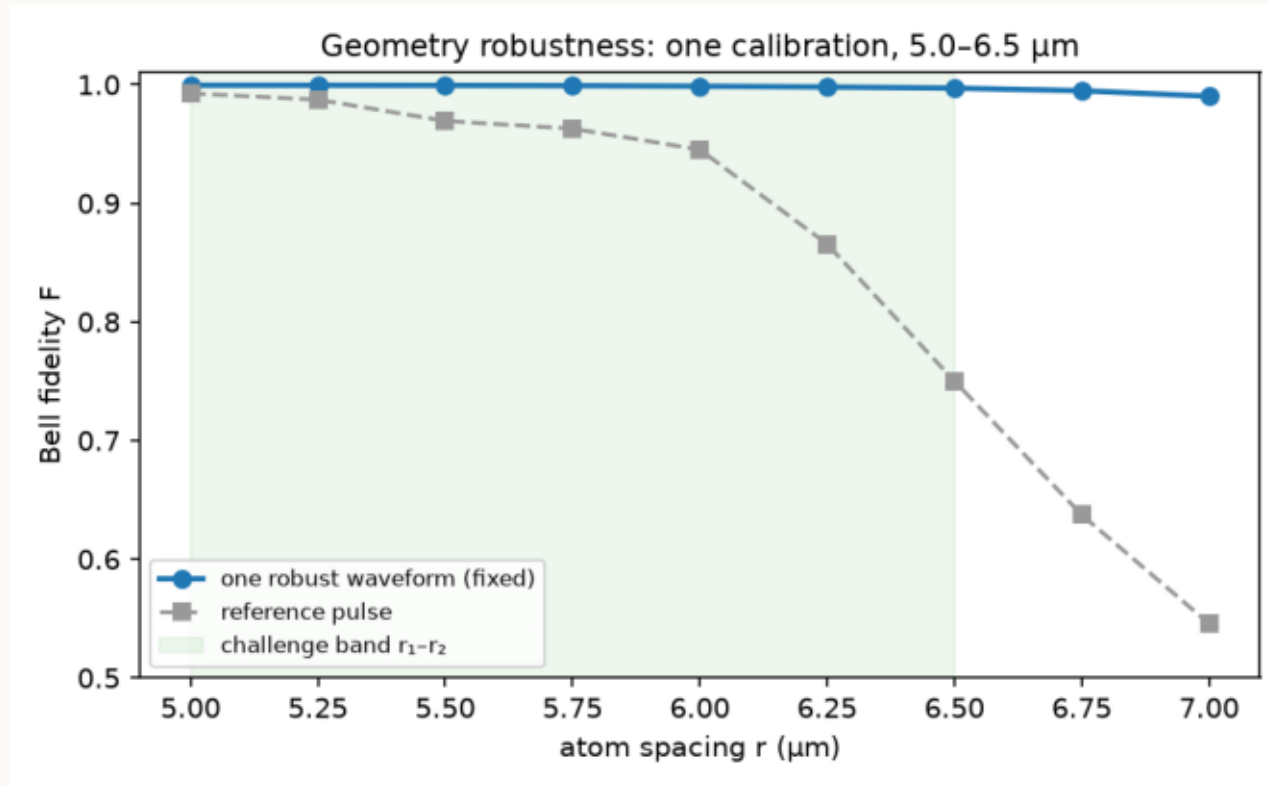
Result: both spacings beaten, validated on Pasqal Cloud

Every pulse validated by the device's own contract checker, then 500 shots each on Pasqal Cloud.



top: our pulse shapes (x: time; blue: power Ω , red: frequency δ) · bottom-left: fidelity bars vs baseline · bottom-right: cloud shots (dots) vs simulation (circles)

Bonus: one pulse that tolerates 30% spacing error



x-axis: atom spacing r (μm) · y-axis: fidelity F of ONE fixed pulse

Real machines place atoms imperfectly.

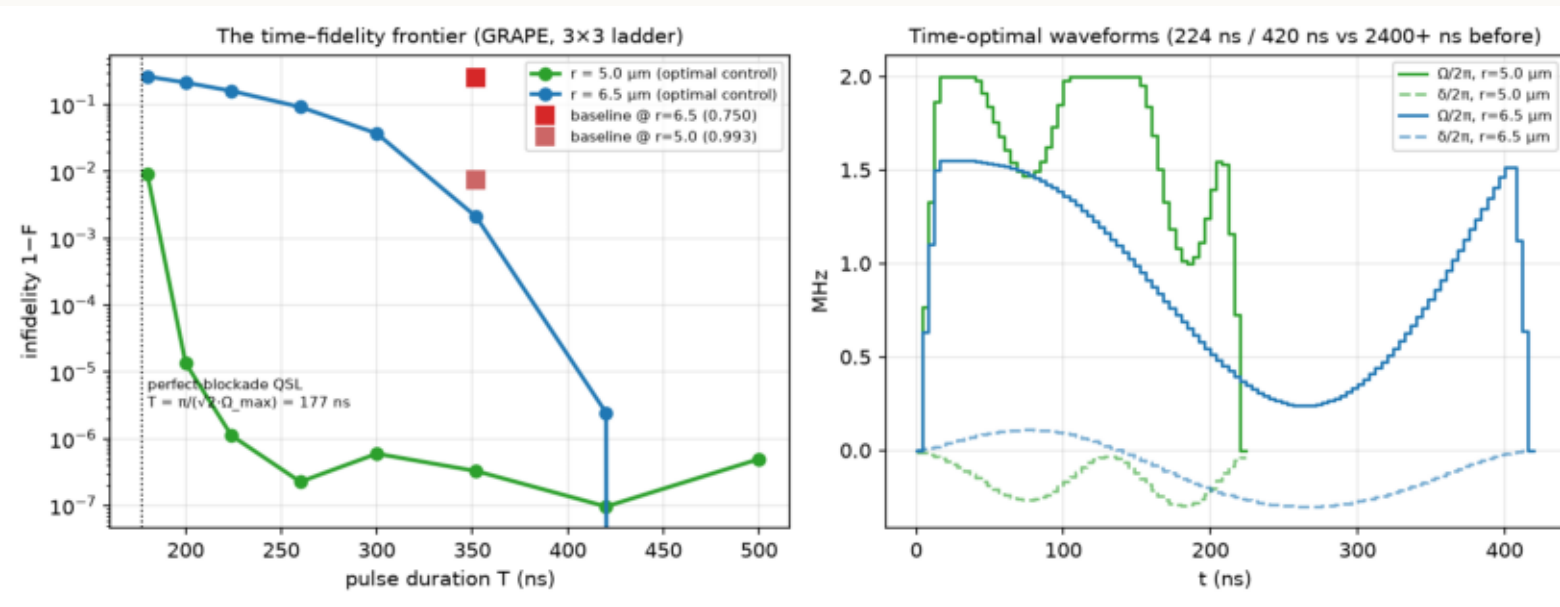
We optimized a single waveform for the worst case over the whole band.

Blue: our fixed pulse holds $F > 0.997$ from 5.0 to 6.5 μm .

Grey: the baseline collapses to 0.55.

One calibration instead of one per geometry — that is the product.

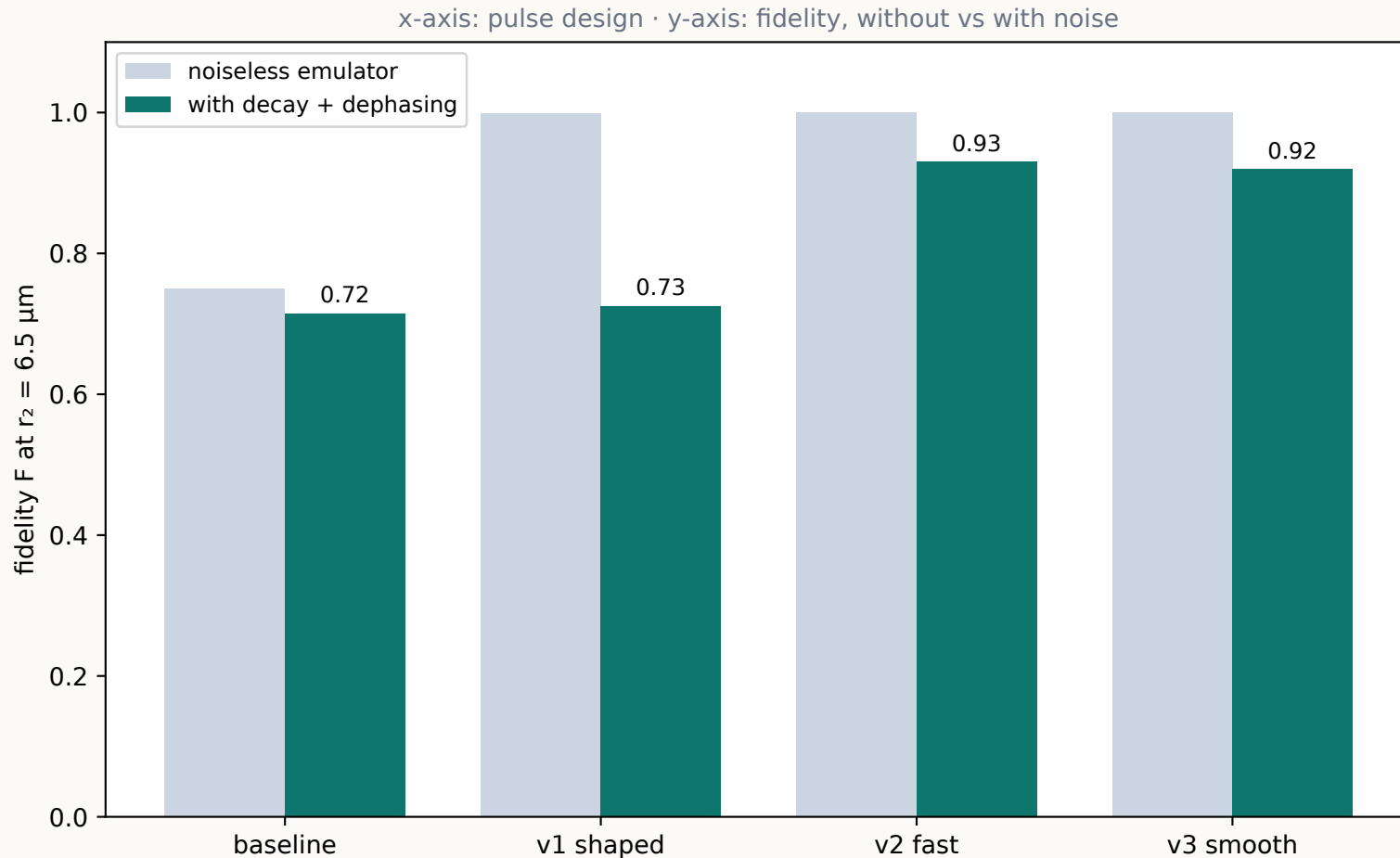
Then we asked: how fast can the best pulse be?



left — x-axis: pulse duration (ns) · y-axis: error $1-F$ (log) · right — the winning waveforms

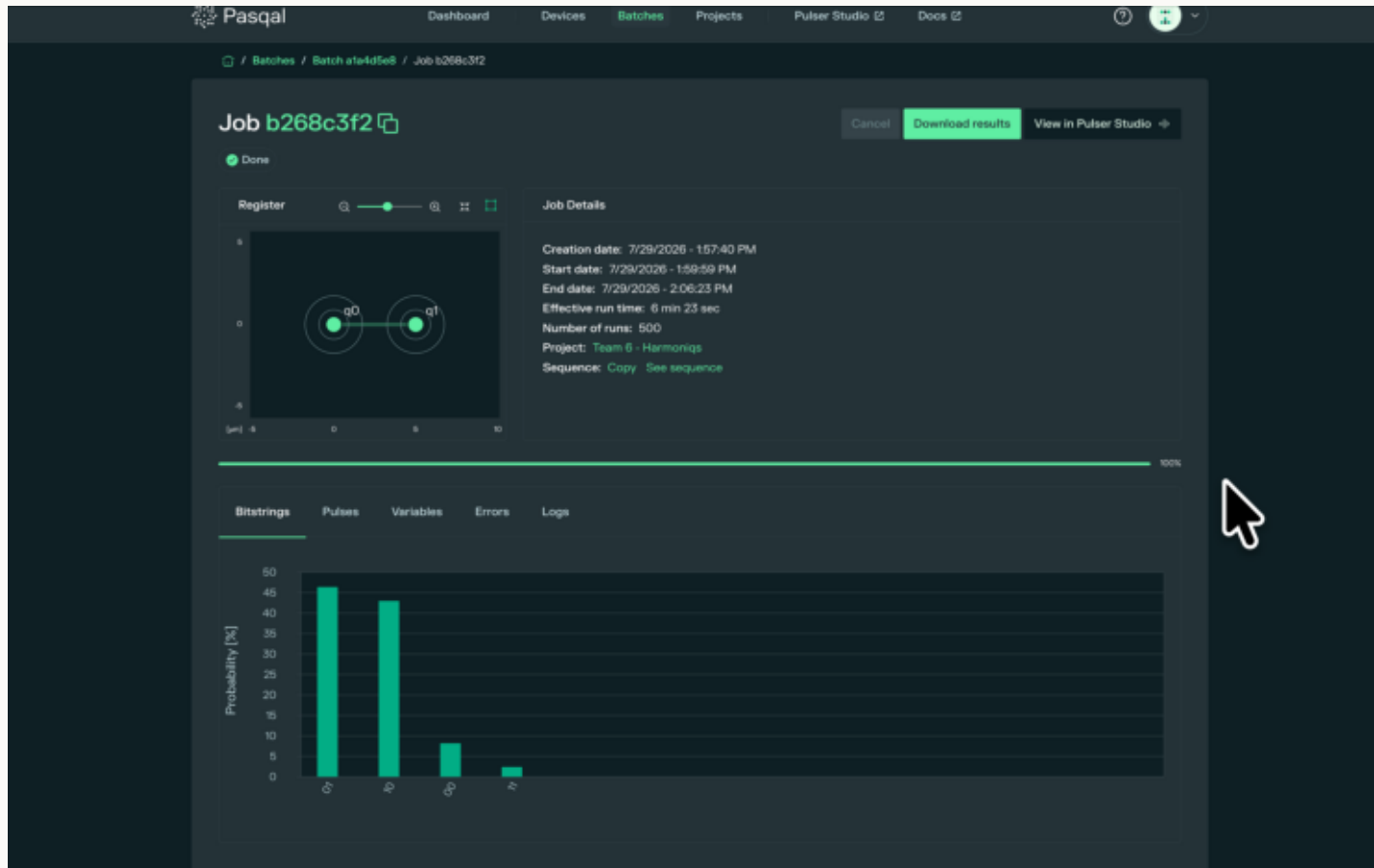
- Theory bound: $T = \pi/(\sqrt{2} \cdot \Omega_{\text{max}}) = 177 \text{ ns}$. No pulse can be faster.
- We reach $F = 0.999999$ at 224 ns (r_1) — 25 ns above the bound.
- At r_2 the weak blockade sets its own wall near 420 ns — a measured speed limit, itself a result.
- Whole frontier: 5.4 s of laptop compute (exact 3-state model + analytic gradients).

Under real noise, the ranking flips — shorter wins



- Dephasing accumulates with time: the 2.7- μs pulse loses 0.27 of F.
- The 420-ns pulse keeps 0.93.
- Design law: pulse duration IS the noise coupling. Minimize T first.
- This is why we sent the short, smooth pulse to the real machine.

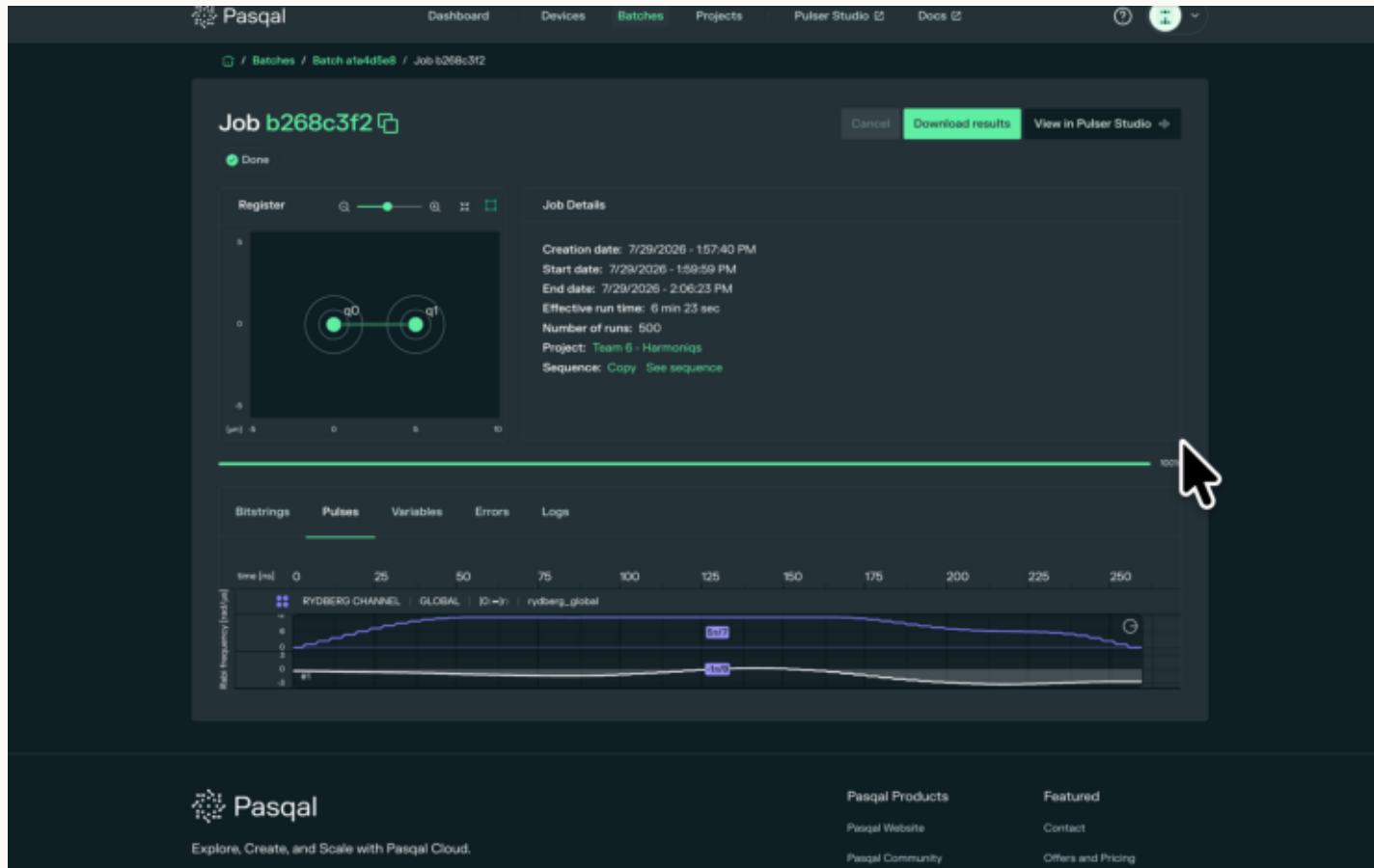
500 shots on the real machine: 89.4% entangled



- The two tall bars ARE the Bell state: '01' 46% + '10' 43% = 89.4% — exactly one atom excited.
- They are equal (within noise): the symmetry physics demands, delivered.
- '11' (blockade violation): 2.4% — forbidden stayed forbidden.
- '00' 8%: readout/decay, the channel our noise model predicted.
- Pre-registered window: hit.

Pasqal's own portal · x-axis: measured two-atom outcome · y-axis: % of 500 shots

The machine's dashboard shows our physics back to us



- 260 nanoseconds, start to finish.
- Power rises smoothly from zero — no jumps (the hardware-true 'v3' pulse our scientists asked for).
- The portal labels the pulse area: $5\pi/7 \approx 0.714\pi$.
- A two-atom collective π -pulse needs area $\pi/\sqrt{2} \approx 0.707\pi$.
- The $\sqrt{2}$ collective enhancement, on the machine's own UI, to 1%.

Pasqal portal, Pulses tab · x-axis: time (ns) · purple: laser power $\Omega(t)$ · white: detuning $\delta(t)$

How the score is computed — no magic

$$F = |\langle \Psi^+ | \psi(T) \rangle|^2$$

- $\psi(T)$ — the state our pulse actually produced (4 numbers for 2 atoms; the simulator computes them exactly).
- $\langle \Psi^+ | \dots \rangle$ — overlap with the perfect Bell state: 'how much of the ideal is in what we made?'
- $|\dots|^2$ — square it to get a probability. 1.000 = perfect, 0 = nothing.
- On hardware there is no ψ to peek at — so the machine takes 500 photographs and we compare the four outcome percentages against the simulation. Agreement within $\sqrt{p(1-p)/500} \approx \pm 2\%$ = validated.

1.000000

our best F, both spacings (simulation)

89.4%

entangled shots on the real machine (500 photos)

Why this matters beyond the score

- Entangling pairs is THE primitive: every neutral-atom algorithm, sensor, or network starts here.
- Our robust pulse holds $F > 0.997$ across a 30% spacing error — real machines misplace atoms, and a pulse that doesn't care means ONE calibration instead of one per geometry. That is uptime.
- Our fast pulses run 6-11× shorter — less exposure to noise means more of the error budget left for the actual computation. Duration is money on quantum hardware.
- The whole design loop is seconds on a laptop — cheap enough to re-run at every calibration cycle.

Built on the Pasqal stack:

- Pulser (open source) — the exact device model + the simulator the judges score with.
- Pasqal Cloud free emulator — 500-shot validation of every pulse before spending hardware budget.
- FRESNEL_CAN1 QPU — the real-atom runs; the portal's register/pulse/bitstring views made every result inspectable by anyone (see the screenshots in this deck).

Challenge 01 — the score sheet

	$r_1 = 5.0 \mu\text{m}$	$r_2 = 6.5 \mu\text{m}$	duration
Baseline (starter pulse)	0.9926	0.7500	352 ns
v1 · simple analytic (4 params)	0.99990	0.99944	~2400 ns
v1r · one spacing-robust pulse	0.99944	0.99701	2400 ns
v2 · time-optimal (at the QSL)	0.999999	0.999998	224 / 420 ns
v3 · hardware-true smooth	1.000000	1.000000	352 / 600 ns

- Cloud: 7 pulses × 500 shots — every population within shot noise of prediction.
- Real QPU: $P_{\text{bell}} = 0.894$, inside the window we published BEFORE the run.
- Every number in this deck regenerates from the public repo in minutes.